Bayesian Small Area Estimation of Under-Five Malaria Prevalence in West Africa

A Preprint

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Abstract

District-level estimates of malaria prevalence are needed to target control interventions, but the household surveys that supply them cover only a fraction of districts and yield unstable estimates wherever few clusters are sampled. We address both problems with a Bayesian spatio-temporal small area estimation framework applied to under-five malaria prevalence across Ghana, Burkina Faso and Côte d'Ivoire, using two rounds of Demographic and Health Survey biomarker data covering 724 districts. Design-based Hájek prevalence estimates and their Taylor-linearised sampling variances enter an area-level Fay-Herriot model equipped with a scaled BYM2 spatial prior and a Knorr-Held Type I space-time interaction. We progress through four specifications: a cross-sectional Ghana baseline, separate country-level space-time models, a single-round pooled three-country model that establishes cross-border smoothing on temporally aligned data, and a pooled space-time model that adds the temporal dimension and is benchmarked against the single-round model to confirm the cross-border structure is not an artefact of joining surveys from different years. The estimated national trajectories differ sharply across the three countries: Ghana and Burkina Faso show large declines between rounds (odds ratios of roughly 0.32 and 0.08), while Côte d'Ivoire shows an increase (odds ratio near 2.1). Urbanisation, proxied by nighttime lights, is the most consistent protective covariate across all three countries. Pooling raises the structured share of spatial variation from roughly one half to four fifths and rescues many noisy districts into a usable reliability range. The framework fills unobserved districts, stabilises noisy direct estimates and flags districts that diverge from their national trend, providing a defensible regional evidence base for subnational malaria planning.

Keywords small area estimation · Bayesian hierarchical models · BYM2 · Fay-Herriot · malaria · spatio-temporal modelling · West Africa

1 Introduction

Malaria continues to impose a heavy burden of childhood illness and death across sub-Saharan Africa, and the countries of West Africa bear a disproportionate share of it (World Health Organization 2023). The interventions that have driven the regional decline of the past two decades insecticide-treated nets, seasonal malaria chemoprevention, intermittent preventive treatment in pregnancy and indoor residual spraying are planned, financed and delivered through district health systems. Decisions about where to send nets, where to run a chemoprevention campaign and where to investigate a resurgence are therefore made at the district scale. The evidence used to make them, however, is usually summarised at the regional or national level, which hides the district-to-district variation that targeting depends on.

Two features of household survey data stand in the way of direct district-level estimation. The first is incomplete spatial coverage. The Demographic and Health Surveys (DHS) provide the most reliable population-based measurements of malaria infection in children through rapid diagnostic testing (Florey 2014; Croft et al. 2018), but survey clusters are placed in only a subset of districts within each country. In the surveys analysed here, between roughly forty and seventy per cent of districts contain at least one cluster in any given round, leaving the remainder with no direct measurement at all. The second feature is sampling instability. Even where a district is sampled, an estimate built from a handful of clusters carries a wide design-based standard error, so that the apparent geography of risk can be dominated by survey noise rather than by any real signal.

Small area estimation responds to both problems by borrowing strength (Rao and Molina 2015; Wakefield et al. 2020). The area-level model of Fay and Herriot treats each district's direct estimate as a noisy reading of an unobserved true prevalence, and separates the sampling error which is treated as known from the survey design from the residual variation that the model is free to learn. A spatial prior lets neighbouring districts inform one another, a temporal component lets survey rounds inform one another, and auxiliary covariates let districts that resemble each other in their environmental and socio-economic profile share information as well. The result is a complete prevalence surface with calibrated uncertainty, including for districts that were never sampled.

This report develops such a framework and applies it jointly to Ghana, Burkina Faso and Côte d'Ivoire. The three countries are geographically contiguous, share the dominant *Anopheles* vector species and the West African monsoon that drives seasonal transmission, and have pursued overlapping but not identical intervention programmes. They are also a useful test of multi-country modelling because their recent prevalence trajectories diverge: as the results below show, two of the three have declined sharply between survey rounds while the third has worsened. We ask four questions. Can a spatial model with a BYM2 prior produce reliable district estimates, including for unobserved districts, from DHS data and standard covariates? How has prevalence moved between rounds in each country, and which districts depart from the national trend? Which covariates drive spatial variation, and do their effects differ across countries? And does a pooled model that smooths across national borders and partially pools covariate effects improve on separate country models while remaining defensible when the survey calendars do not align?

The remainder of the report is organised around the data pipeline and a sequence of four models. Section 2 describes the surveys, the adjacency structure and the covariate panel, and sets out how the direct estimates and their sampling variances are computed. Section 3 states the common modelling machinery. The four models then build in sequence: a cross-sectional baseline that isolates the spatial structure (Section 4), country-level space-time models that recover each national trajectory (Section 5), a single-round pooled model that establishes cross-border smoothing on temporally aligned data (Section 6), and a pooled space-time model that adds the temporal dimension and is checked against the single-round model (Section 7). Section 8 sets out the planned external validation against the Malaria Atlas Project and the sensitivity analyses. Section 9 draws the findings together, Section 10 states the limitations, and Section 11 concludes.

2 Data and Direct Estimation

2.1 Survey data

The data come from the DHS programme, which fields nationally representative household surveys using a stratified two-stage cluster design (Croft et al. 2018). Enumeration areas are selected within strata defined by region and urban or rural residence, and households are selected within each sampled cluster. Children aged six to fifty-nine months are tested for malaria infection by rapid diagnostic test, the field standard since around 2010 (Florey 2014). Cluster coordinates are displaced by up to five kilometres in urban areas and ten in rural areas to protect respondent confidentiality. We use the household member recode file for the individual-level biomarker and the cluster GPS shapefile for spatial referencing. Two rounds are used per country: Ghana 2014 and 2022, Burkina Faso 2010 and 2021, and Côte d'Ivoire 2012 and 2021 (Table 1).

Table 1: DHS survey rounds and district counts by country.

Country	Round 1	Round 2	Districts
Burkina Faso	DHS 2010	DHS 2021	351
Côte d'Ivoire	DHS 2012	DHS 2021	113
Ghana	DHS 2014	DHS 2022	260

2.2 Administrative boundaries and adjacency

District boundaries were taken from the Global Administrative Areas database, at the second administrative level for Ghana and the third for Burkina Faso and Côte d'Ivoire, which correspond to the units at which the national malaria programmes operate. The combined panel contains 724 districts (260, 351 and 113 respectively). Adjacency was defined by queen contiguity using `spdep` (Bivand et al. 2013), and any districts left without neighbours were reconnected using a small snap tolerance so that the intrinsic conditional autoregressive prior remains proper (Freni-Sterrantino et al. 2018). The single-country graphs contain 709, 973 and 296 edges; the combined 724-district graph contains 2,056 edges, of which 78 cross a national border (Table 2). These cross-border edges are what allow districts near a frontier to borrow information from their neighbours in the adjacent country.

Table 2: Adjacency graph sizes and BYM2 scaling factors. The combined graph adds 78 cross-border edges.

Country	Districts	Edges	Scaling factor
Ghana	260	709	4.36
Burkina Faso	351	973	3.36
Côte d'Ivoire	113	296	9.30
Combined	724	2056	NA

2.3 Covariates

Seven covariates were assembled at the district-round level: elevation (SRTM, population-weighted), rainfall (CHIRPS v2.0, September to December mean), nighttime lights (NASA Black Marble VIIRS) as a proxy for urbanisation (Elvidge et al. 2013), temperature (CRU TS v4.09, September to December mean), travel time to the nearest health facility (Weiss et al., population-weighted) (Weiss et al. 2019), log population density (WorldPop), and a multidimensional poverty headcount (OPHI MPI and national census). Each covariate was standardised within country, so that coefficients describe the effect of a one-standard-deviation change, a single weakly informative prior applies across all of them, and the Hamiltonian Monte Carlo sampler explores a better-conditioned posterior. All retained pairwise correlations stayed below the conventional threshold of 0.7 (Dormann et al. 2013); nighttime lights and log population density, though related, were both kept because they capture distinct aspects of settlement.

2.4 Direct estimates and sampling variances

For each observed district i in round t we formed a design-based prevalence estimate using the Hájek ratio (Hájek 1971),

$$\hat{p}^{w}_{it} = \frac{\sum_k d_k \, z_k}{\sum_k d_k}, \tag{1}$$

where z_k is the rapid-test result for child k and d_k the survey weight. Standard errors were obtained by Taylor-series linearisation (Binder 1983) using the `survey` package (Lumley 2010), with the stratified two-stage design carried through the variance calculation. To fit the Gaussian Fay-Herriot likelihood we worked on the logit scale, transforming the standard error by the delta method,

$$y_{it} = \operatorname{logit}(\hat{p}^{w}_{it}), \qquad \widehat{\mathrm{SE}}(y_{it}) = \frac{\widehat{\mathrm{SE}}(\hat{p}^{w}_{it})}{\hat{p}^{w}_{it}\,(1-\hat{p}^{w}_{it})}. \tag{2}$$

For numerical stability, prevalences were bounded to $[0.02, 0.98]$ and logit-scale standard errors to $[0.1, 2.0]$ before entering the model. Coverage varied by country and round: of 260 Ghana districts, 153 were observed in 2014 and 124 in 2022; of 351 Burkina Faso districts, 232 in 2010 and 153 in 2021; and of 113 Côte d'Ivoire districts, 80 in 2012 and 103 in 2021.

3 Common Methodology

All four models share a Fay-Herriot likelihood, a scaled BYM2 spatial prior and a common set of penalised-complexity priors. We state these once here; each model section then adds only what is specific to it.

3.1 Fay-Herriot likelihood

The direct estimate is modelled as a noisy observation of a latent logit prevalence,

$$y_{it} \sim \mathcal{N}\big(\theta_{it},\, \psi_{it}^2\big)\,, \tag{3}$$

with the sampling variance ψ_{it}^2 treated as known (Wakefield et al. 2020). This is the standard area-level simplification; its consequences for districts with very few clusters are returned to in Section 10.

3.2 BYM2 spatial prior

Every model places a scaled BYM2 prior (Riebler et al. 2016) on the district-level spatial effect, written in non-centred form,

$$b_i = \sigma_b\Big(\sqrt{1-\phi}\, v_i + \sqrt{\phi/s}\, u_i\Big)\,, \qquad v_i \sim \mathcal{N}(0,1), \qquad u_i \sim \mathrm{ICAR}(W). \tag{4}$$

Here v_i is an unstructured component, u_i a spatially structured intrinsic conditional autoregressive component, σ_b the marginal standard deviation of the combined effect, and $\phi \in (0,1)$ the share of variance that is spatially structured. The scaling constant s, computed from the graph, makes ϕ comparable across the different adjacency structures, which the original BYM model could not guarantee (Besag et al. 1991). The non-centred form avoids the funnel geometry that otherwise couples σ_b to the random effects.

3.3 Priors

The intercept used a weakly informative $\mathcal{N}(-1.85, 1)$ prior, centred near the logit of a moderate prevalence. The spatial standard deviation used an exponential penalised-complexity prior approximating $P(\sigma_b > 1) = 0.01$ (Simpson et al. 2017), the mixing parameter a Beta$(3, 2)$ prior with a mild preference for structured over unstructured variation, and standardised covariate coefficients a shared $\mathcal{N}(0, 1)$ prior. The space-time interaction standard deviation and the between-country slope standard deviations both used Exp(2) penalised-complexity priors.

3.4 Computation

All models were fitted in Stan (Stan Development Team 2024) through the `cmdstanr` interface, using the No-U-Turn Sampler with non-centred parameterisations throughout. Each model ran four chains with 1,500 warmup and 2,000 sampling iterations. The cross-sectional and country-level space-time models used `adapt_delta` 0.99 and a maximum tree depth of 14; the larger pooled models used 0.97 and 13. Convergence was judged by the $\hat{R}$ statistic, bulk and tail effective sample sizes (Vehtari et al. 2017) and inspection of trace, rank and pairs plots; across the models reported here $\hat{R}$ stayed at or below about 1.02 with no divergent transitions. Competing specifications were compared by approximate leave-one-out cross-validation (Vehtari et al. 2017).

4 Model 1: Cross-Sectional Ghana Baseline

4.1 Specification

The baseline model isolates the spatial component by restricting attention to a single country and round, Ghana in 2022,

$$\theta_i = \mu + b_i + \mathbf{x}_i^\top \boldsymbol{\beta}, \tag{5}$$

with b_i the scaled BYM2 effect defined in the common methodology. The attraction of the BYM2 parameterisation is that it does not force a choice between an unstructured and a structured spatial model; it contains both as limiting cases and lets the data decide the balance between them. Recall that

$$b_i = \sigma_b \Big(\sqrt{1-\phi}\, v_i + \sqrt{\phi/s}\, u_i \Big), \tag{6}$$

where v_i is the unstructured (independent) component and u_i the structured ICAR component. Setting the mixing parameter $\phi = 0$ removes the structured part entirely and the model collapses to a purely independent (IID) specification in which each district is shrunk toward the national mean on its own; setting $\phi = 1$ removes the unstructured part and the model becomes a pure ICAR in which every district is informed wholly by its neighbours. For intermediate values, ϕ is interpretable as the proportion of the residual spatial variance that is geographically structured, so the single parameter both decides whether IID or ICAR behaviour dominates and quantifies the split between them. To confirm that this flexibility is warranted rather than assumed, we fit the two limiting cases explicitly IID (ϕ fixed at 0) and ICAR (ϕ fixed at 1) and compare all three by leave-one-out cross-validation.

4.2 Results

The cross-validation is decisive about what to exclude and permissive about what to keep (Figure 1, Table 3). The pure ICAR model is clearly the worst of the three, trailing the best by about seven units of expected log predictive density with a standard error that does not cover zero; forcing every district to be explained entirely by its neighbours is too rigid for Ghana's data. The BYM2 and IID models, by contrast, are statistically indistinguishable, separated by less than one predictive unit, which tells us that a large share of the district-level variation in a single cross-section is idiosyncratic rather than smoothly geographic. The BYM2 posterior makes the same point from the inside: it places the structured share at $\phi \approx 0.44$, so the data themselves attribute roughly half of the residual spatial variance to geography and half to district-specific noise.

We adopt the BYM2 prior as the working spatial structure for this and every subsequent model, for two reasons that the cross-validation makes concrete. First, BYM2 is never worse than the better of its two limiting cases here, because it can reproduce either by moving ϕ; choosing it costs nothing in predictive terms relative to IID while retaining the ability to express structure where structure exists. Second, and decisively for an estimation task whose purpose is to fill unobserved districts, only the structured component allows an unsampled district to borrow from its neighbours. A pure IID model would shrink every unobserved district to the same national mean and so could not produce a spatially coherent surface, which is precisely what subnational targeting requires. Letting ϕ be estimated rather than fixed means each later model the country-level space-time models and the two pooled models inherits a spatial prior that adapts the IID-to-ICAR balance to its own data rather than committing to one extreme in advance. The BYM2 fit for Ghana 2022 gives a national intercept of -1.82 on the logit scale, a marginal spatial standard deviation of 0.88, and an implied national prevalence near 14 per cent.

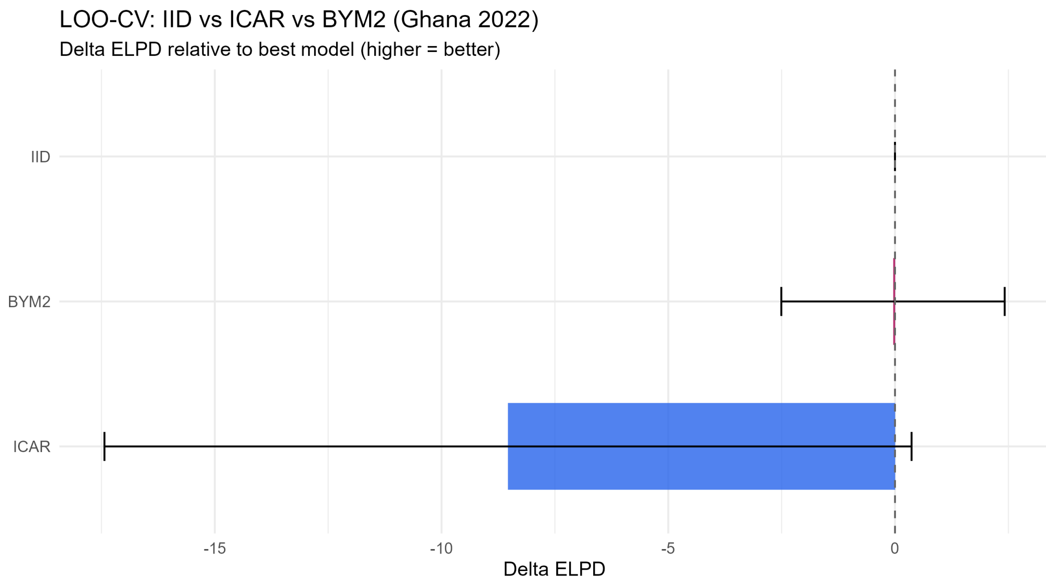

Figure 1: Leave-one-out cross-validation for the Ghana 2022 cross-section: change in expected log predictive density relative to the best model, with standard errors. The pure ICAR model is clearly worse; BYM2 and IID are within noise of each other, so BYM2 is retained because it spans both and supports spatial borrowing.

Table 3: Ghana 2022 cross-section: leave-one-out comparison. BYM2 and the unstructured model are within noise of each other; the pure ICAR model is clearly worse.

Model	ELPD diff	SE diff	p_loo
BYM2	0.0	0.0	68.8
IID	-0.3	1.3	69.3
ICAR	-7.3	3.8	72.8

The baseline confirms that the BYM2 machinery fills the unobserved districts and produces a coherent surface, but it can say nothing about change over time. The next model adds the temporal dimension.

5 Model 2: Country-Level Space-Time Models

5.1 Specification

Fitting each country separately, the latent prevalence gains a national temporal shift and an unstructured space-time interaction,

$$\theta_{it} = \mu^{(c)} + b_i^{(c)} + \delta_t^{(c)} + \gamma_{it}^{(c)} + \mathbf{x}_{it}^\top \boldsymbol{\beta}^{(c)}, \qquad \gamma_{it}^{(c)} \sim \mathcal{N}(0, \sigma_\gamma^2). \tag{7}$$

With only two rounds, the Knorr-Held Type I interaction is the only identifiable choice: a temporally structured prior over two points collapses to a single increment, and spatially structured interactions are not separable from the main spatial effect at this length of series (Knorr-Held 2000; Goicoa et al. 2018). The first round is set as the reference ($\delta_1 = 0$) and the second estimated freely.

5.2 Results

The three countries tell strikingly different stories (Table 4). In Ghana, the temporal shift is $\delta_2 = -1.14$, corresponding to a fall in modelled national prevalence from about 30.5 to 12.3 per cent. In Burkina Faso the decline is far steeper, with $\delta_2 = -2.52$ and prevalence dropping from roughly 71.2 to 16.8 per cent. Côte d'Ivoire moves in the opposite direction: $\delta_2 = +0.78$, with prevalence rising from about 17.4 to 31.3 per cent.

The contrast between two countries in rapid retreat and a third moving against the regional tide is the central empirical finding, and the next sections show it to be robust to how the data are pooled.

The spatial parameters are also informative. The structured share ϕ ranges from 0.47 in Ghana through 0.59 in Côte d'Ivoire to 0.64 in Burkina Faso, so that between a half and two thirds of the unexplained variation is geographic in each country. Burkina Faso carries the largest marginal spatial standard deviation at 1.04, consistent with its pronounced north-to-south transmission gradient. The interaction standard deviation sits near 0.5 in all three countries, indicating that individual districts do depart from the national trend by a meaningful amount; the term γ_{it} records exactly which ones.

Table 4: Country-level space-time posterior summaries. Ghana and Burkina Faso decline sharply between rounds; Côte d'Ivoire rises.

Parameter	Ghana	Burkina Faso	Côte d'Ivoire
δ_2 (time shift)	-1.14	-2.52	+0.78
Round 1 prevalence	30.5\%	71.2\%	17.4\%
Round 2 prevalence	12.3\%	16.8\%	31.3\%
σ_b	0.72	1.04	0.82
ϕ	0.47	0.64	0.59
σ_γ	0.52	0.52	0.48

Mapping the interaction term for Ghana identifies districts that have diverged from the national decline: a band of districts whose 95 per cent intervals exclude zero have fallen behind the national pace of improvement, while others have outpaced it. These are the districts a programme manager would single out for investigation, since they are not explained by the country-wide trend, by their neighbours or by the measured covariates.

What the country-level models cannot do is share information across borders or pull a regional consensus from the covariate effects. Côte d'Ivoire in particular, with the fewest districts, would benefit from borrowing. The pooled models that follow address this, beginning with the temporally aligned single-round model.

6 Model 3: Single-Round Pooled Three-Country Model

6.1 Motivation

The country-level models treat each country as a spatial island, yet the three countries share borders, vector species and the monsoon that drives transmission. The natural next step is to pool them on a single adjacency graph so that districts near a frontier can borrow from their neighbours across the national line. Pooling across two survey rounds at once, however, raises a question about the spatial smoother: the Round 1 surveys span four calendar years (Burkina Faso 2010, Côte d'Ivoire 2012, Ghana 2014), and a graph that smooths residuals across borders in that earlier round would be mixing space with time over a period in which intervention coverage and rainfall changed appreciably. It is not obvious that cross-border adjacency is meaningful when the surfaces being joined are years apart.

We therefore establish the cross-border case first on its cleanest footing, using only the contemporaneous second round (Burkina Faso 2021, Côte d'Ivoire 2021, Ghana 2022), which falls within a single year. With a single round the temporal shift and the space-time interaction disappear, and the model reduces to a pure spatial BYM2 over all 724 districts with hierarchical covariate slopes. The cross-border smoothing now operates on genuinely simultaneous surfaces, so the spatial borrowing cannot be confounded with temporal change. This single-round pooled model is the methodologically conservative cross-border specification; the temporal extension that follows in Section 7 is then read against it.

6.2 Specification

For each observed Round 2 district i,

$$\theta_i = \mu + \alpha_{c[i]} + b_i + \mathbf{x}_i^\top \boldsymbol{\beta}_{c[i]}, \qquad \beta_{ck} \sim \mathcal{N}(\bar{\beta}_k, \tau_k^2). \tag{8}$$

The country deviations α_c are constrained to sum to zero so that they are identified against the grand intercept μ, the spatial effect b_i is the BYM2 field on the combined 724-district graph including its 78 cross-border edges, and the covariate slopes for each country are drawn around a shared regional mean $\bar{\beta}_k$ with between-country spread τ_k. The closest published cross-border application used a comparable two-year alignment window (Saldarriaga 2020); the single-year window here is tighter still. Priors and the adjacency graph are exactly those of the common methodology, so nothing in the comparison with the temporal model that follows is driven by a change of specification.

6.3 Results

7 Model 4: Pooled Three-Country Space-Time Model

7.1 Motivation

The single-round pooled model of Section 6 establishes the cross-border spatial structure on temporally aligned data but discards half the evidence: it says nothing about change between rounds. The full space-time model restores the temporal dimension, fitting all 724 districts and both rounds jointly with country intercepts, country-specific temporal shifts and hierarchical covariate slopes. In doing so it must confront the concern raised earlier that the cross-border edges in the earlier round smooth across surfaces several years apart. The model's defence is structural: the country-specific shifts $\delta_{c,t}$ absorb each country's own level and trend before the shared spatial field operates on the residuals, so the field is smoothing deviations from each national trajectory rather than raw prevalence across time. Whether that defence holds is then checked empirically by comparing this model's spatial parameters and covariate means against the temporally aligned single-round model of Section 6.

7.2 Specification

$$\theta_{it} = \mu + \alpha_{c[i]} + b_i + \delta_{c[i],t} + \gamma_{it} + \mathbf{x}_{it}^{\top}\boldsymbol{\beta}_{c[i]}, \qquad \beta_{ck} \sim \mathcal{N}(\bar{\beta}_k, \tau_k^2). \tag{9}$$

The country deviations sum to zero, the temporal shifts $\delta_{c,t}$ free each national trajectory with the first round as reference, and γ_{it} is the Knorr-Held Type I interaction. The spatial effect operates on the same combined graph as Model 4, so the only additions relative to that model are the temporal terms.

7.3 Results

Table 5: Pooled space-time model: country-specific temporal shifts and implied between-round odds ratios. The shared spatial field has $\sigma_b = 0.65$ and $\phi = 0.83$.

Country	δ_2	Odds ratio
Ghana	-1.13	0.32
Côte d'Ivoire	+0.74	2.13
Burkina Faso	-2.53	0.08

8 External Validation and Sensitivity Analyses

The estimates presented above are internally coherent and well behaved under the model's own diagnostics, but a full assessment requires checking them against an independent source and probing their sensitivity to the modelling choices. To be done...

8.1 Validation against the Malaria Atlas Project

The Malaria Atlas Project publishes model-based geostatistical surfaces of *Plasmodium falciparum* parasite-rate for sub-Saharan Africa, built from a different evidence base and a different modelling tradition than the area-level approach used here (Bhatt et al. 2015; Weiss et al. 2019). To be done.....

8.2 Sensitivity analyses

9 Discussion

10 Limitations and Future Work

11 Conclusion

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